

Projects

00103335: Deep Learning and Reinforcement Learning

Fall 2026

Objective. Use simulation to study two phenomena: double descent in ridgeless least squares and uncertainty quantification for averaged stochastic gradient descent (SGD). Your submission must be reproducible: state all random seeds, use double precision, and label every axis and curve.

Problem 1. Double-descent experiment

Generate data from

$$y = X\beta + \varepsilon, \quad X_{ij} \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1), \quad \varepsilon \sim \mathcal{N}(0, I_n), \quad n = 200, \quad \beta = \sqrt{5} e_1,$$

and vary the number of features over

$$p \in \{20, 40, 60, 80, 100, 120, 140, 160, 180, 199, 200, 220, 240, 260, 300, 400, 600\}.$$

For each value of p , fit the minimum-norm least-squares estimator $\hat{\beta} = X^+y$. Generate an independent test matrix $X_{\text{test}} \in \mathbb{R}^{2000 \times p}$ with i.i.d. standard Gaussian entries, and estimate the prediction risk by

$$R = \frac{1}{2000} \left\| X_{\text{test}} \hat{\beta} - X_{\text{test}} \beta \right\|_2^2.$$

- Repeat the experiment 20 times independently for every p , recording R in each repetition.
- Make two separate plots of R against p : one showing the median and one showing the sample mean. Use each plot to identify the value of p at which the peak occurs.
- Repeat the same experiment, including both plots, for $\beta = \sqrt{2} e_1$ and $\beta = e_1$. Use the same values of p and the same number of repetitions.

Problem 2. Asymptotically valid inference for averaged SGD

Let

$$\sigma(u) = \frac{1}{1 + e^{-u}}, \quad \Sigma_X = \begin{pmatrix} 1 & 0.4 \\ 0.4 & 1 \end{pmatrix}, \quad \theta^* = \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

At iteration t , draw a fresh observation, independently over time,

$$X_t \sim \mathcal{N}_2(0, \Sigma_X), \quad Y_t | X_t \sim \text{Bernoulli}(\sigma(X_t^\top \theta^*)).$$

For $z = (x, y)$, use the two-dimensional logistic loss

$$\ell(\theta; z) = \log(1 + e^{x^\top \theta}) - y x^\top \theta, \quad L(\theta) = \mathbb{E}\{\ell(\theta; Z)\}.$$

The inferential target is the scalar contrast

$$\psi^* = c^\top \theta^* = -\frac{1}{\sqrt{5}}, \quad c = \frac{1}{\sqrt{5}} \begin{pmatrix} 1 \\ 2 \end{pmatrix}.$$

- (a) Show that the stochastic gradient and per-observation Hessian are

$$g(\theta; x, y) = \{\sigma(x^\top \theta) - y\}x, \quad A(\theta; x) = \sigma(x^\top \theta)\{1 - \sigma(x^\top \theta)\}xx^\top.$$

Explain why θ^* is the unique minimizer of L .

- (b) Starting from $\theta_0 = (0, 0)^\top$, run the one-pass recursion

$$\theta_t = \theta_{t-1} - \eta_t g(\theta_{t-1}; X_t, Y_t), \quad \eta_t = 1.6(t + 10)^{-0.6}.$$

Fix the burn-in at $t_0 = 200$. For $T > t_0$, define

$$I_T = \{t_0 + 1, \dots, T\}, \quad n_T = T - t_0, \quad \bar{\theta}_T = \frac{1}{n_T} \sum_{t \in I_T} \theta_t, \quad \hat{\psi}_T = c^\top \bar{\theta}_T.$$

Use fresh data at every step; do not cycle through a fixed data set. Using this averaged estimator, construct each of the following nominal 95% confidence intervals.

Plug-in. With $g_t = g(\theta_{t-1}; X_t, Y_t)$, compute

$$\hat{H}_T = \frac{1}{n_T} \sum_{t \in I_T} A(\theta_{t-1}; X_t), \quad \hat{S}_T = \frac{1}{n_T} \sum_{t \in I_T} g_t g_t^\top, \quad \hat{V}_T = \hat{H}_T^{-1} \hat{S}_T \hat{H}_T^{-1}.$$

Use the full sandwich form, even though the information identity holds in the correctly specified population model. The interval is

$$\mathcal{I}_T^{\text{PI}} = \left[\hat{\psi}_T \pm 1.96 \sqrt{\frac{c^\top \hat{V}_T c}{n_T}} \right].$$

Online multiplier bootstrap. Maintain $B = 200$ perturbed paths. Independently draw $U_t^{(b)} \stackrel{\text{iid}}{\sim} \text{Exp}(1)$, independently of the data, and update

$$\theta_0^{*(b)} = \theta_0, \quad \theta_t^{*(b)} = \theta_{t-1}^{*(b)} - \eta_t U_t^{(b)} g(\theta_{t-1}^{*(b)}; X_t, Y_t), \quad b = 1, \dots, B.$$

All bootstrap paths in an outer replication must use the same observed stream $\{(X_t, Y_t)\}$; only their multipliers differ. Define

$$\bar{\theta}_T^{*(b)} = \frac{1}{n_T} \sum_{t \in I_T} \theta_t^{*(b)}, \quad D_T^{*(b)} = \sqrt{n_T} c^\top (\bar{\theta}_T^{*(b)} - \bar{\theta}_T).$$

If $q_{p,T}^*$ is the empirical p -quantile of the B values $D_T^{*(b)}$, use the basic bootstrap interval

$$\mathcal{I}_T^{\text{Boot}} = \left[\hat{\psi}_T - \frac{q_{0.975,T}^*}{\sqrt{n_T}}, \hat{\psi}_T - \frac{q_{0.025,T}^*}{\sqrt{n_T}} \right].$$

Random scaling. Relabel the retained iterates as $\vartheta_j = \theta_{t_0+j}$, $j = 1, \dots, n_T$, and let

$$\bar{\vartheta}_s = \frac{1}{s} \sum_{j=1}^s \vartheta_j, \quad \hat{\psi}_s = c^\top \bar{\vartheta}_s, \quad \hat{r}_T = \frac{1}{n_T^2} \sum_{s=1}^{n_T} s^2 (\hat{\psi}_s - \hat{\psi}_T)^2.$$

Use

$$\mathcal{I}_T^{\text{RS}} = \left[\hat{\psi}_T \pm 6.747 \sqrt{\frac{\hat{r}_T}{n_T}} \right].$$

The value 6.747 is the 0.975 critical value of the scalar random-scaling limit distribution.

(c) Repeat the complete experiment independently $R = 500$ times, using

$$T \in \{500, 1000, 2000, 5000, 10000\}.$$

Within one outer replication, run a single data stream and its B bootstrap paths to $T = 10000$, recording all five checkpoints. For $m \in \{\text{PI}, \text{Boot}, \text{RS}\}$, estimate the empirical coverage rate

$$\hat{C}_m(T) = \frac{1}{500} \sum_{r=1}^{500} \mathbf{1}\{\psi^* \in \mathcal{I}_{T,r}^m\}.$$

Make one figure whose horizontal axis is the number of iterations T and whose vertical axis is empirical coverage rate. Plot one labeled curve per method, a horizontal line at 0.95, and the Monte Carlo reference band

$$0.95 \pm 1.96 \sqrt{\frac{0.95(0.05)}{500}} \approx [0.931, 0.969].$$

Also report the mean interval length for each method at $T = 10000$.

- References.** T. Hastie, A. Montanari, S. Rosset, and R. J. Tibshirani, *Surprises in high-dimensional ridgeless least squares interpolation*, The Annals of Statistics, 50(2):949–986, 2022. doi:[10.1214/21-AOS2133](https://doi.org/10.1214/21-AOS2133).
- Y. Fang, J. Xu, and L. Yang, *Online bootstrap confidence intervals for the stochastic gradient descent estimator*, Journal of Machine Learning Research, 19(78):1–21, 2018. [jmlr.org/papers/v19/17-370.html](https://jmlr.org/papers/v19/fang18.html).
- S. Lee, Y. Liao, M. H. Seo, and Y. Shin, *Fast and robust online inference with stochastic gradient descent via random scaling*, Proceedings of the AAAI Conference on Artificial Intelligence, 36(7):7381–7389, 2022. doi:[10.1609/aaai.v36i7.20701](https://doi.org/10.1609/aaai.v36i7.20701).